Multimodal Education Creator

Report

Course Name: GENERATIVE AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: 04

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Academic Year: 2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives:-

1. Problem Statement:-

Traditional learning platforms provide static content and lack personalization. Students often struggle to get context-aware explanations and visual learning support. Moreover, most systems cannot dynamically adapt to new knowledge sources like custom documents.

2. Project Objectives:-

- To build an AI-powered educational content generator
- To generate simple explanations for any topic using LLM
- To provide visual learning support (images)
- To implement Retrieval-Augmented Generation (RAG) for accurate answers
- To allow users to upload PDF documents dynamically
- To improve responses using user-provided knowledge sources

3. Scope of the Project:-

- Supports topic-based learning queries
- Allows PDF-based knowledge expansion
- Works for educational content generation
- Can be extended for:
    - e-learning platform
    - AI tutors
    - Document-based Q&A systems

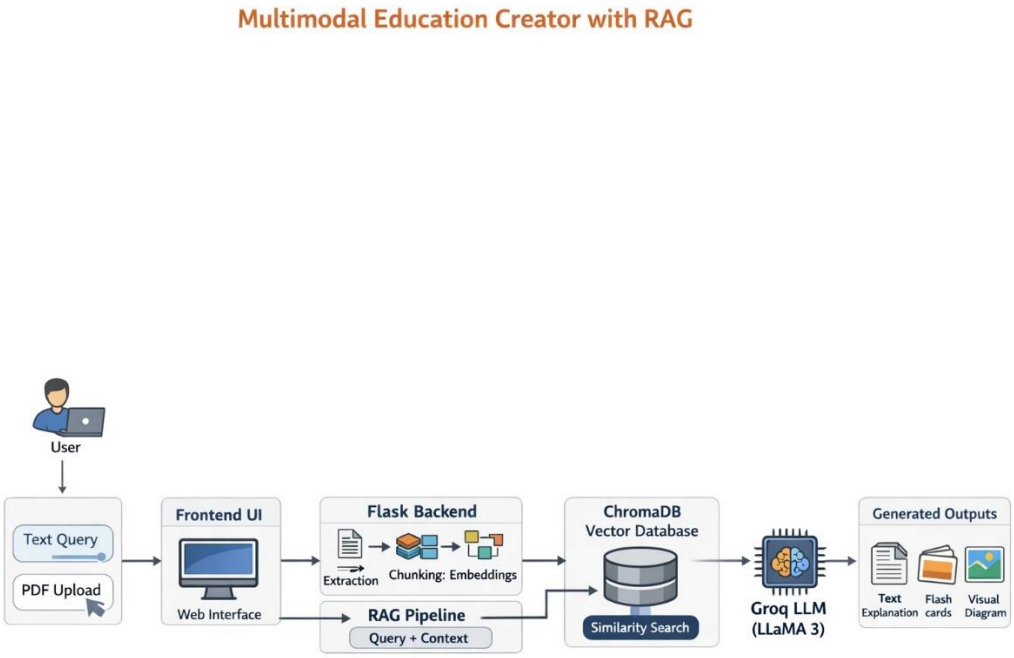
Proposed Solution:-

1. Key Features

- Topic-based explanation generation
- Flashcard generation (conceptual support)
- Image generation using Wikipedia API

- RAG-based contextual answering
- PDF upload interface for dynamic knowledge addition
- Login authentication system

2. Overall Architecture / Workflow:-



3. Tools & Technologies Used:-

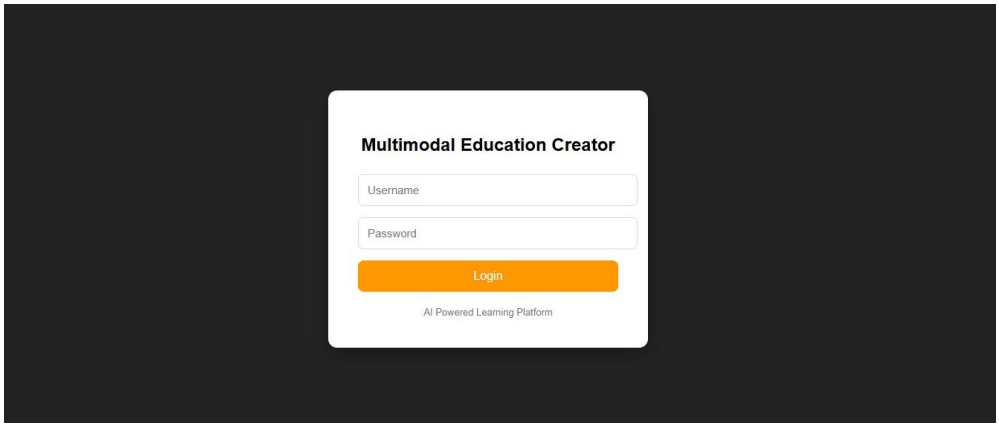
Technology	Purpose
• Flask	Backend web framework
• HTML/CSS	Frontend UI
• Groq API (LLaMA 3)	LLM for text generation

- LangChain — RAG pipeline orchestration
- ChromaDB — Vector database
- HuggingFace Embeddings — Text → vector conversion
- PyPDF — PDF text extraction
- Wikipedia API — Image generation

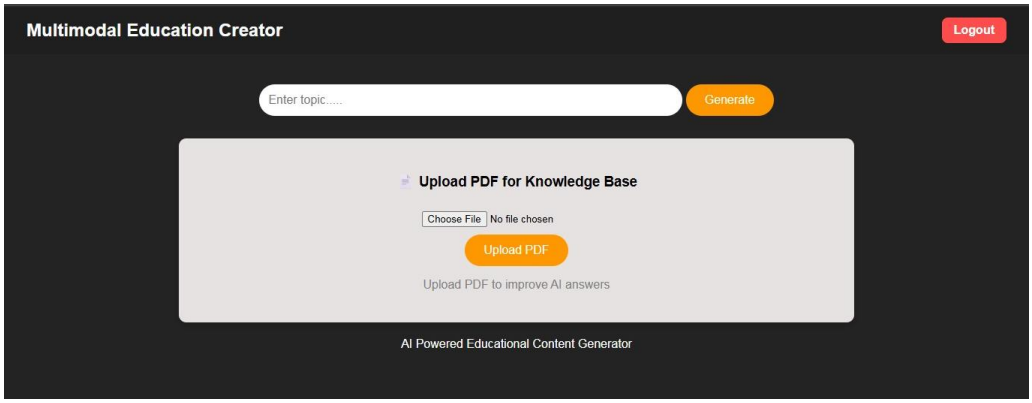
Results & Output:-

1. Screenshots / Outputs:-

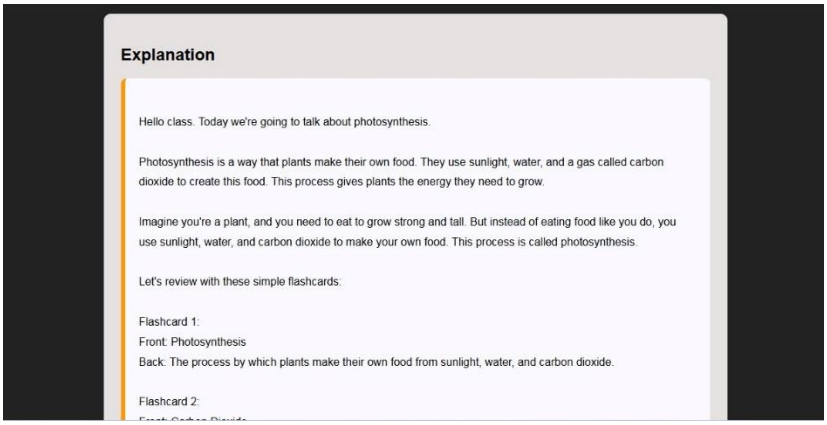
- Login page:-



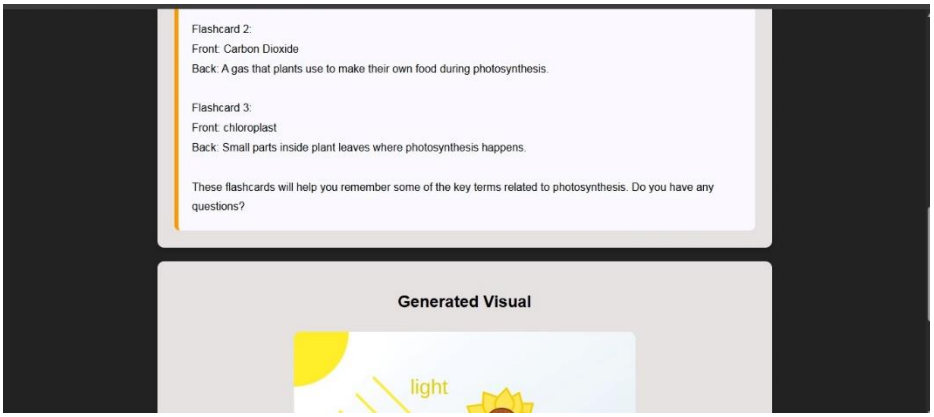
- Home page (topic input + PDF upload)



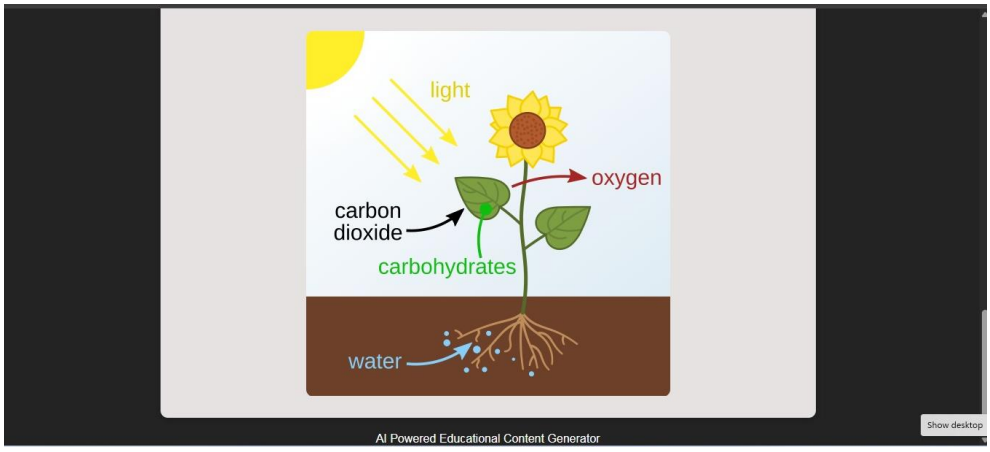
- Generated explanation output



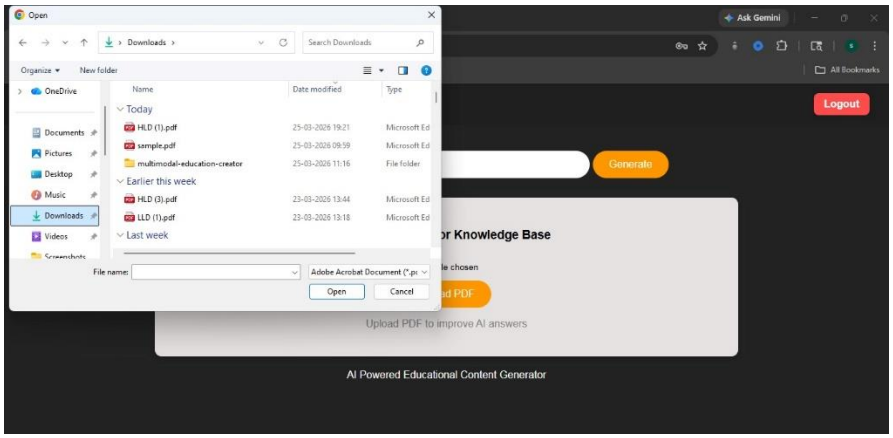
- Flashcards section



- Image output



- PDF upload interface



2. Reports / Models:-

- Embedding model: all-MiniLM-L6-v2
- Vector DB: ChromaDB storing chunked text
- LLM: LLaMA 3 via Groq API
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3. Key Outcomes:-

- Successfully implemented RAG pipeline
- Enabled dynamic knowledge ingestion via PDF
- Improved response accuracy using context retrieval
- Integrated LLM + Vector DB + UI
- Built a full-stack AI application

Conclusion:-

- This project successfully demonstrates the integration of Generative AI with Retrieval-Augmented Generation (RAG) to create a smart educational assistant.

- By combining LLM capabilities with vector-based retrieval, the system generates more accurate and context-aware explanations.

- The addition of a PDF upload interface enhances the system's flexibility, allowing users to expand the knowledge base dynamically.

- Overall, the project provides a strong foundation for AI-driven learning systems.

Future Scope & Enhancements:-

- Multi-PDF upload support
- Drag-and-drop file upload UI
- Flashcard parsing and structured display
- User-specific knowledge storage
- Real-time streaming responses
- Deployment on cloud (AWS / GCP)
- Chat-based conversational interface
- Support for multiple file formats (DOCX, TXT)